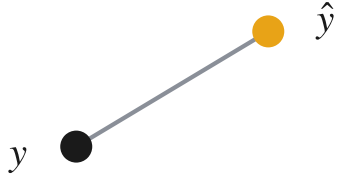
Accuracy

prediction vs. ground truth



$$D(y, \hat{y}) = 0$$

the prediction equals the ground truth

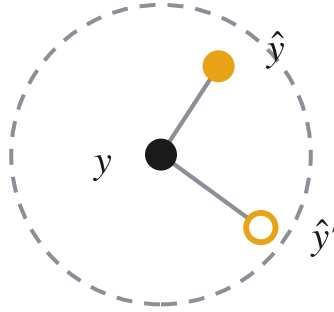
take $a' = a$

reduces to

Accurate Fairness

both, bounded by one constraint

$$K \cdot d((x, a), (x, a'))$$



$$D(y, f(x, a')) \leq K \cdot d((x, a), (x, a'))$$

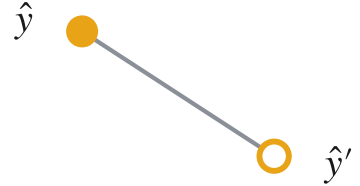
for every similar $(x, a') \in I(x, a)$

Theorem 1

implies

Individual Fairness

prediction vs. similar counterpart



$$D(\hat{y}, \hat{y}') \leq K \cdot d((x, a), (x, a'))$$

similar individuals get similar predictions

y ground truth $\hat{y}$ prediction, (x, a) $\hat{y}'$ prediction, counterpart (x, a')